

Linear Regression with One Regressor

Chapter 4

ECON3500: Econometrics and Applications

Spring 2026

Learning Roadmap

! The Central Question

How do we move from observing **relationships** in data to making **predictions** and testing **causal claims**?

By the end of this chapter, you will be able to:

Set up and estimate a simple linear regression model

Interpret slopes and intercepts with precision

Calculate and understand fitted values and residuals

Assess model quality using R^2 , TSS, ESS, SSR, and SER

State the key assumptions behind OLS estimation

Implement everything in Stata

Our Journey

The linear regression model

Estimating coefficients: ordinary least squares

Measuring how well the model fits

Assumptions for causal interpretation

Sampling distributions and uncertainty

Common misconceptions and clarifications

Implementing OLS in Stata

Bringing it all together

The Linear Regression Model

Motivating Question

Returns to education

What are the economic returns to education?

Why this matters:

We observe that more educated people tend to earn more

Is this just correlation, or something more?

How much more do we expect someone with 16 years of education to earn vs. someone with 12 years?

Can we **quantify** the relationship systematically?

Our goal: Move from vague observations to a **prediction rule** we can compute from data.

Data Source

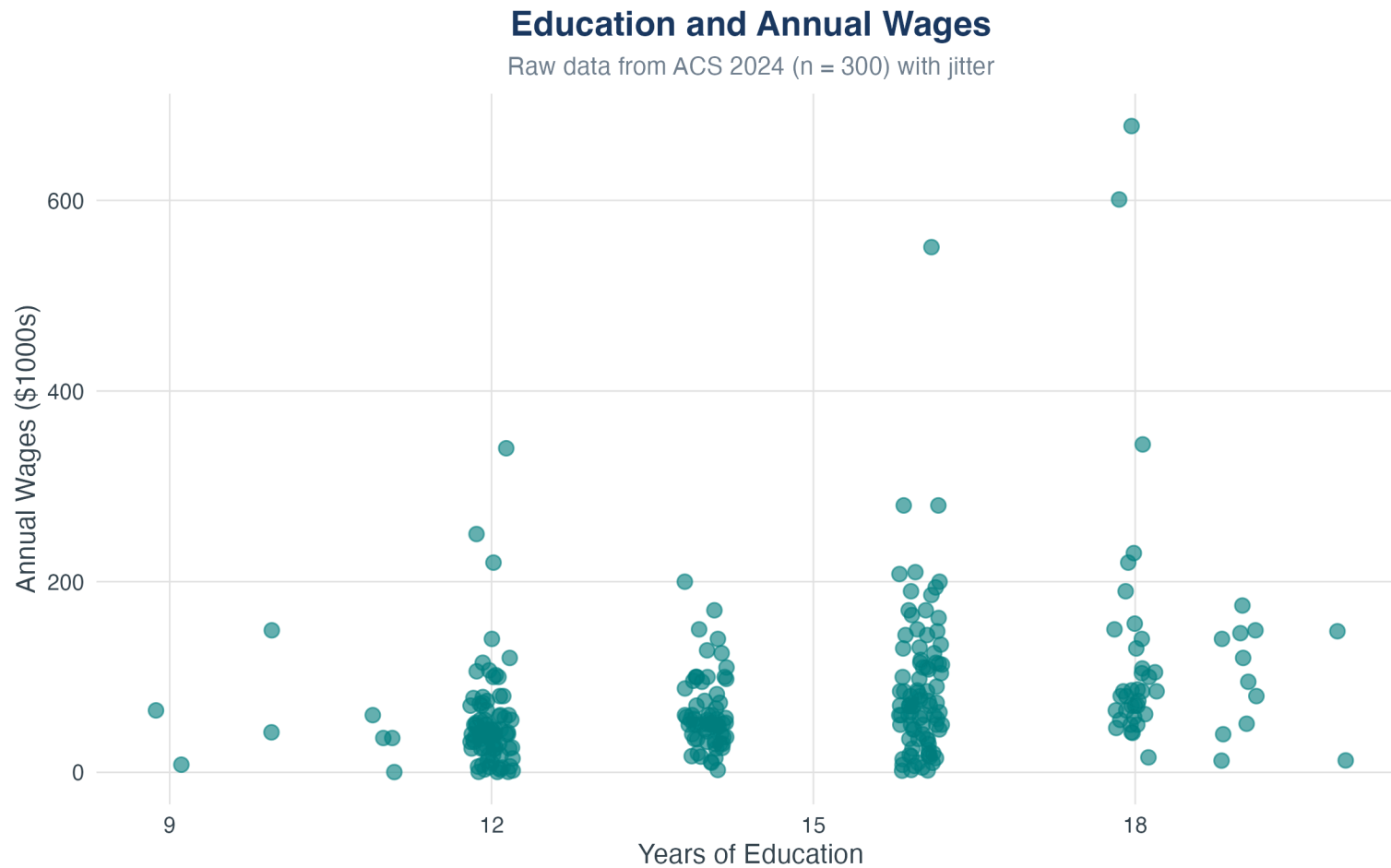
Example data: Education and Annual Wages

Source: American Community Survey (ACS) 2024 **Sample:** 300 observations

Variables: **Education:** Years of schooling (8-20 years) **Wages:** Annual earnings in thousands of 2024 dollars

Note: Based on realistic patterns from ACS microdata

First Look at the Data



What do you see?

From Picture to Model

Three tools for describing relationships:

Scatter plots give us a first look

Correlation tells us direction and strength

Linear regression gives us a mathematical summary we can use for prediction

! Key Insight

Regression transforms a cloud of points into an equation. That equation becomes a **model** we can interrogate, test, and use.

What Regression Analysis Does

Regression helps us:

Predict: Given someone's education, what wages would we expect?

Explain: How does a one-year change in education map to wage changes?

Control: What's the education-wages relationship *holding other factors fixed*?

Vocabulary:

Dependent variable (Y): the outcome we want to explain (wages)

Independent variable (X): the driver we use to explain it (education)

The Population Model

In the population, the true relationship is:

$$Y = \beta_0 + \beta_1 X + u$$

Components:

β_0 : **intercept** — baseline level of Y when $X = 0$

β_1 : **slope** — change in Y from a one-unit increase in X

u : **error term** — all other influences on Y not captured by X

► This is a *model* — a simplified representation of reality. Our job: estimate β_0 and β_1 from data.

Understanding the Error Term

$$\text{Wages} = \beta_0 + \beta_1 \cdot \text{Education} + u$$

What's in u ?

Ability and talent

Work experience

Gender

Family background

Pure luck

Measurement error

Critical Point

We can't observe u , but its properties determine whether we can interpret β_1 **causally**. More on this later.

Ceteris Paribus: Holding Other Things Fixed

Ceteris paribus = “other things equal”

$$Y = \beta_0 + \beta_1 X + u$$

Taking changes:

$$\Delta Y = \beta_1 \Delta X + \Delta u$$

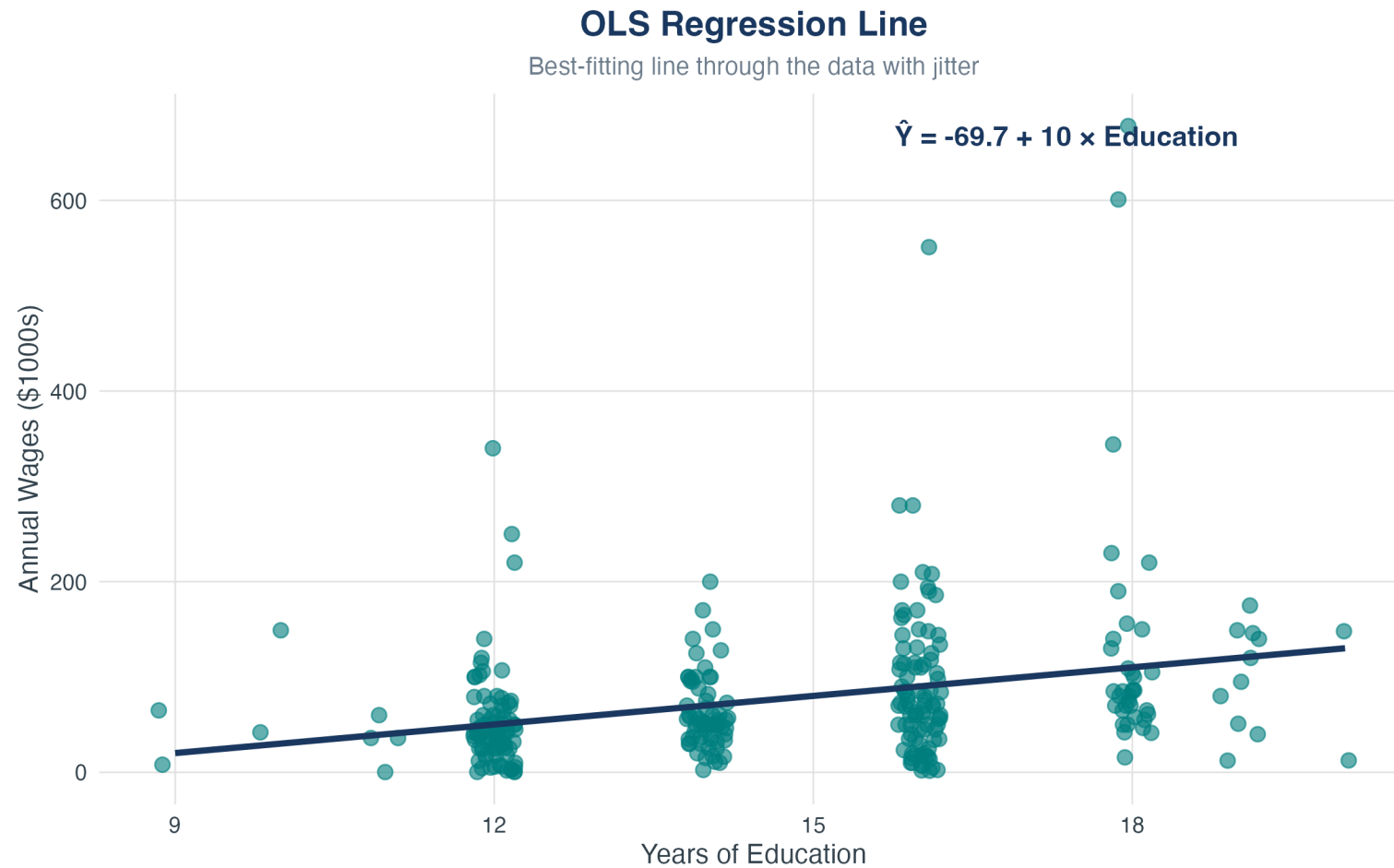
Holding u fixed means $\Delta u = 0$:

$$\Delta Y = \beta_1 \Delta X$$

i Slope Interpretation

β_1 is the change in Y associated with a one-unit change in X , **holding all other factors (captured in u) fixed**.

The Regression Line in Action



The line is: $\hat{Y} = \beta_0 + \beta_1 X$

Estimating Coefficients: Ordinary Least Squares

From Population to Sample

The challenge:

In the population: $Y_i = \beta_0 + \beta_1 X_i + u_i$

We observe (X_i, Y_i) for $i = 1, \dots, n$

We do **not** observe u_i

We must *estimate* β_0 and β_1 from the sample

Notation:

$\hat{\beta}_0, \hat{\beta}_1$ = our estimates (the “hats” indicate estimates)

Goal: Make these estimates as close as possible to the true β_0, β_1

The OLS Criterion

Idea: Choose $\hat{\beta}_0$ and $\hat{\beta}_1$ to make prediction errors as small as possible.

For each observation, the prediction error is:

$$Y_i - (\beta_0 + \beta_1 X_i)$$

OLS minimizes the sum of squared errors:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n [Y_i - (\beta_0 + \beta_1 X_i)]^2$$

Why square?

Positive and negative errors don't cancel out

Penalizes large mistakes more than small ones

Leads to a closed-form solution (calculus magic!)

Deriving the OLS Estimators [Optional]

For Math Lovers

This derivation uses calculus to show where the OLS formulas come from. Feel free to skip if you prefer!

Setup: Minimize the sum of squared residuals

$$S(\beta_0, \beta_1) = \sum_{i=1}^n [Y_i - (\beta_0 + \beta_1 X_i)]^2$$

Step 1: Take first-order conditions (FOCs)

$$\frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^n [Y_i - (\beta_0 + \beta_1 X_i)] = 0$$

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^n X_i [Y_i - (\beta_0 + \beta_1 X_i)] = 0$$

Deriving the OLS Estimators (continued)

Step 3: Substitute $\hat{\beta}_0$ into the second FOC

$$\begin{aligned}\sum X_i Y_i &= \hat{\beta}_0 \sum X_i + \hat{\beta}_1 \sum X_i^2 \\ \sum X_i Y_i &= (\bar{Y} - \hat{\beta}_1 \bar{X}) \sum X_i + \hat{\beta}_1 \sum X_i^2 \\ \sum X_i Y_i &= \bar{Y} n \bar{X} - \hat{\beta}_1 n \bar{X}^2 + \hat{\beta}_1 \sum X_i^2\end{aligned}$$

Step 4: Solve for $\hat{\beta}_1$

$$\begin{aligned}\hat{\beta}_1 (\sum X_i^2 - n \bar{X}^2) &= \sum X_i Y_i - n \bar{X} \bar{Y} \\ \hat{\beta}_1 &= \frac{\sum X_i Y_i - n \bar{X} \bar{Y}}{\sum X_i^2 - n \bar{X}^2} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}\end{aligned}$$

Result: This is the formula for $\hat{\beta}_1$ in “The OLS Formulas” slide!

The OLS Formulas

Solution from calculus (see previous slides for derivation):

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

Intuition:

$\hat{\beta}_1$ depends on how X and Y covary

Need variation in X to estimate β_1 (if all X_i are the same, we can't learn about the slope!)

The line always passes through $(\bar{X}, \bar{Y})$

Fitted Values and Residuals

Once we have $\hat{\beta}_0$ and $\hat{\beta}_1$:

i Fitted Value

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

This is the model's prediction for observation i .

i Residual

$$\hat{u}_i = Y_i - \hat{Y}_i$$

This is the prediction error for observation i .

► **Key property:** $\sum_{i=1}^n \hat{u}_i = 0$ (residuals always sum to zero)

Visualizing Residuals

Stata: Running OLS Regression

Basic command:

```
regress wages education
```

Output includes:

Coefficient estimates: $\hat{\beta}_0$ (_cons) and $\hat{\beta}_1$ (education)

Standard errors (for inference - we'll cover this later)

R² (measure of fit)

Number of observations

Generate fitted values and residuals:

```
predict wages_fitted, xb  
predict residual, residuals
```

Reading Stata Regression Output

Understanding the table:

```
Source | SS df MS Number of obs = 300
-----+----- F(1, 298) = 54.60
Model | 298054 1 298054 Prob > F = 0.0000
Residual | 1626752 298 5459.23 R-squared = 0.155
-----+----- Adj R-squared = 0.152
Total | 1924806 299 6436.81 Root MSE = 73.883

-----+-----
wages | Coef. Std. Err. t P>|t| [95% Conf. Interval]
-----+-----
education | 13.022 1.762 7.39 0.000 9.552 16.492
_cons | -114.740 25.633 -4.48 0.000 -165.187 -64.292
-----+-----
```

Coef. column: $\hat{\beta}_1 = 13.02$, $\hat{\beta}_0 = -114.7$

R-squared: 0.155 (model explains 15.5% of variation)

Root MSE: Standard error of regression (SER) = 73.88

Std. Err.: We'll use this for hypothesis tests (next chapter!)

Interpreting the Output

Example results:

$$\widehat{\text{Wages}} = -114.7 + 13.02 \cdot \text{Education}, \quad N = 300$$

Interpretations:

Slope (13.02): Each additional year of education is associated with approximately \$13,020 more in annual wages

Intercept (-114.7): Predicted wages when education = 0 (not meaningful here!)

Predictions:

- Person with 16 years of education: $\hat{Y} = -114.7 + 13.02(16) = 93.6\text{k}$
- Person with 12 years of education: $\hat{Y} = -114.7 + 13.02(12) = 41.5\text{k}$

Residual example: If someone with 12 years of education earns \$35k, their residual is $40 - 41.5 = -6.5\text{k}$

Measuring How Well the Model Fits

Decomposing Variation

For any observation:

$$Y_i = \hat{Y}_i + \hat{u}_i$$

We can split total variation into two parts:

$$\underbrace{Y_i - \bar{Y}}_{\text{Total deviation}} = \underbrace{\hat{Y}_i - \bar{Y}}_{\text{Explained deviation}} + \underbrace{\hat{u}_i}_{\text{Unexplained deviation}}$$

Squaring and summing across all observations:

$$\underbrace{\sum_{i=1}^n (Y_i - \bar{Y})^2}_{\text{TSS}} = \underbrace{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}_{\text{ESS}} + \underbrace{\sum_{i=1}^n \hat{u}_i^2}_{\text{SSR}}$$

The Sum of Squares Trinity

i Total Sum of Squares (TSS)

$$TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

Measures total variation in Y around its mean.

i Explained Sum of Squares (ESS)

$$ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$$

Variation in Y explained by the model (variation in fitted values).

i Sum of Squared Residuals (SSR)

$$SSR = \sum_{i=1}^n u_i^2$$

Variation in Y **not** explained by the model (variation in residuals).

Why Does $TSS = ESS + SSR$? (The Algebra)

Start with the decomposition:

$$Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + (Y_i - \hat{Y}_i)$$

Square both sides:

$$(Y_i - \bar{Y})^2 = [(\hat{Y}_i - \bar{Y}) + (Y_i - \hat{Y}_i)]^2$$

Expand:

$$(Y_i - \bar{Y})^2 = (\hat{Y}_i - \bar{Y})^2 + (Y_i - \hat{Y}_i)^2 + 2(\hat{Y}_i - \bar{Y})(Y_i - \hat{Y}_i)$$

Sum over all i :

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + 2 \sum_{i=1}^n (\hat{Y}_i - \bar{Y})(Y_i - \hat{Y}_i)$$

The cross-product term equals **ZERO!** (by properties of OLS)

$$TSS = ESS + SSR$$

Visualizing the Decomposition

R-Squared: The Fraction Explained

 R^2 (R-squared)

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{SSR}{TSS}$$

Properties:

$$0 \leq R^2 \leq 1$$

$R^2 = 0$: The model explains none of the variation (horizontal line at $\bar{Y}$)

$R^2 = 1$: The model explains all variation (perfect fit, all points on the line)

Typical values: $R^2 = 0.10$ to 0.30 in social sciences (many factors matter!)

Critical Warning

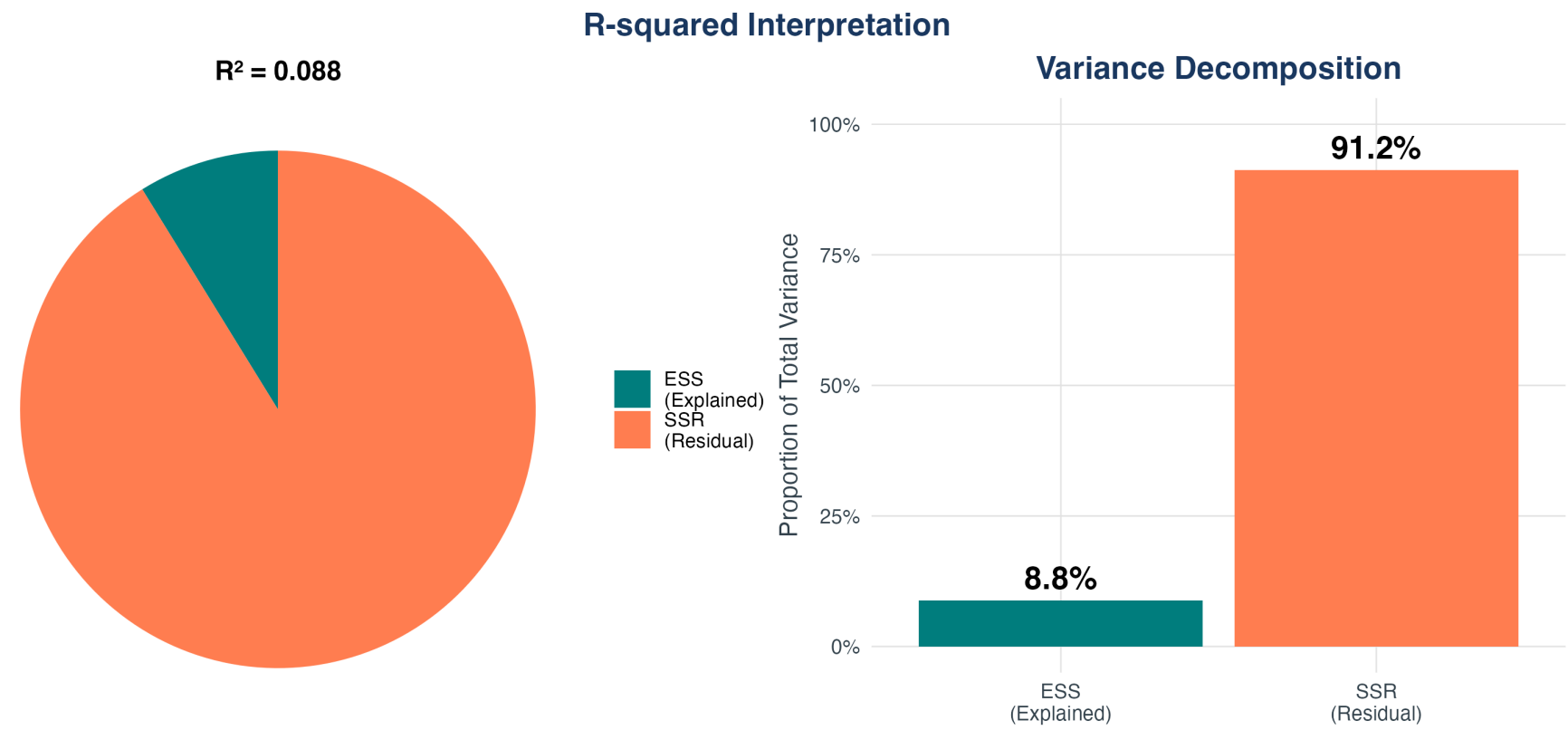
A high R^2 does **not** mean:

The relationship is causal

The model is correctly specified

You have the “right” model

R-Squared Visualization



Standard Error of the Regression (SER)

How big are the typical residuals?

 Standard Error of Regression

$$SER = \sqrt{\frac{SSR}{n-2}} = \sqrt{\frac{\sum_{i=1}^n u_i^2}{n-2}}$$

Also called the “root mean squared error” (RMSE).

Interpretation:

Measures typical size of prediction errors

In original units of Y (unlike R^2 which is unitless)

Lower SER = better fit

Divide by $n - 2$ (not n) because we estimated 2 parameters

Example: If $SER = 7.9$ in our education-wages regression, typical prediction error is about \$7,900.

Assumptions for Causal Interpretation

A Cautionary Tale: High $R^2 \neq$ Causality

Provocative Example

Suppose we regress **drowning deaths** on **ice cream sales**:

$$\widehat{\text{Drownings}} = 2.1 + 0.003 \cdot \text{Ice Cream Sales}$$

$R^2 = 0.89$ (very high fit!)

Does ice cream *cause* drowning?

► **No!** Both are caused by a third factor: **summer weather**

Hot days $\Rightarrow$ more ice cream sales

Hot days $\Rightarrow$ more swimming $\Rightarrow$ more drownings

The lesson: A model can fit the data well (high R^2) but still fail to identify a causal effect. This is why we need **assumptions**.

Unbiasedness: Getting the right answer

! The Fundamental Question

We've estimated $\hat{\beta}_1$, but is it correct? is $\hat{\beta}_1$ an **unbiased** estimator of β_1 ?

Three least squares assumption

Three key assumptions for unbiased OLS:

Zero conditional mean: $E[u_i | X_i] = 0$

- Holds in RCT setting—we try to approximate this
- Same as saying that u_i and X_i are uncorrelated . . .

i.i.d. sampling: (X_i, Y_i) are independent and identically distributed

Finite fourth moments: Large outliers are unlikely (finite kurtosis)

Assumption 1: Zero Conditional Mean

i Zero Conditional Mean

$$E[u_i | X_i] = 0$$

The expected value of the error term is zero, **after conditioning on X** .

Both X and u have distributions in the population

If, for example, $X = \textit{education}$, then we could (in principle) figure out its distribution in the population of adults

Suppose (for simplicity), that u is ability (or age or gender or marital status, etc.). u will also have a distribution in the population

So we are restricting how u and X relate to each other in the population

Simplifying assumption: $E[u] = 0$

First, we make a simplifying assumption without loss of generality:

$$E[u] = 0$$

where $E[\cdot]$ is the expected value operator

Normalizing ability to be zero in the population should be harmless. It is.

Adjusting the intercept

The presence of β_0 in

$$Y = \beta_0 + \beta_1 X + u$$

allows us to assume that $E[u] = 0$. If the average of u is different from zero, then we could just adjust the intercept, leaving the slope the same.

If $\alpha_0 = E[u]$, then we can just add and subtract:

$$Y = (\beta_0 + \alpha_0) + \beta_1 X + (u - \alpha_0)$$

New intercept is $\beta_0 + \alpha_0$ Easy! But, our slope is unchanged.

Definition of simple regression model

KEY QUESTION: How do we need to restrict the dependence between u and X ?

We could assume that u and x are uncorrelated in the population:

$$\text{Corr}(X, u) = 0$$

Zero correlation works for many purposes, but it only implies that u and X are not **linearly** related. Ruling out only linear dependence can cause problems with interpretation and makes analysis more difficult.

Definition of simple regression model

A better assumption involves the mean of the error term for each “slice” of the population determined by the values of X :

$$E[u|X] = E[u] \quad \text{for all values of } X$$

Where $E[u|X]$ is “the expected value of u given X .”

We say that u is **mean independent** of x .

How realistic is this?

Definition of simple regression model

Suppose that u is “ability” and X is years of education. Then we need, for example:

$$E[ability|X = 8] = E[ability|X = 12] = E[ability|X = 16]$$

The average ability is the same in different portions of the population with an 8th grade education, 12th grad education, and four-year college education.

Zero conditional mean assumption.

When we combine this assumption with our normalization ($E[u] = 0$), then we get

$$E[u|X] = 0 \quad \text{for all values of } X$$

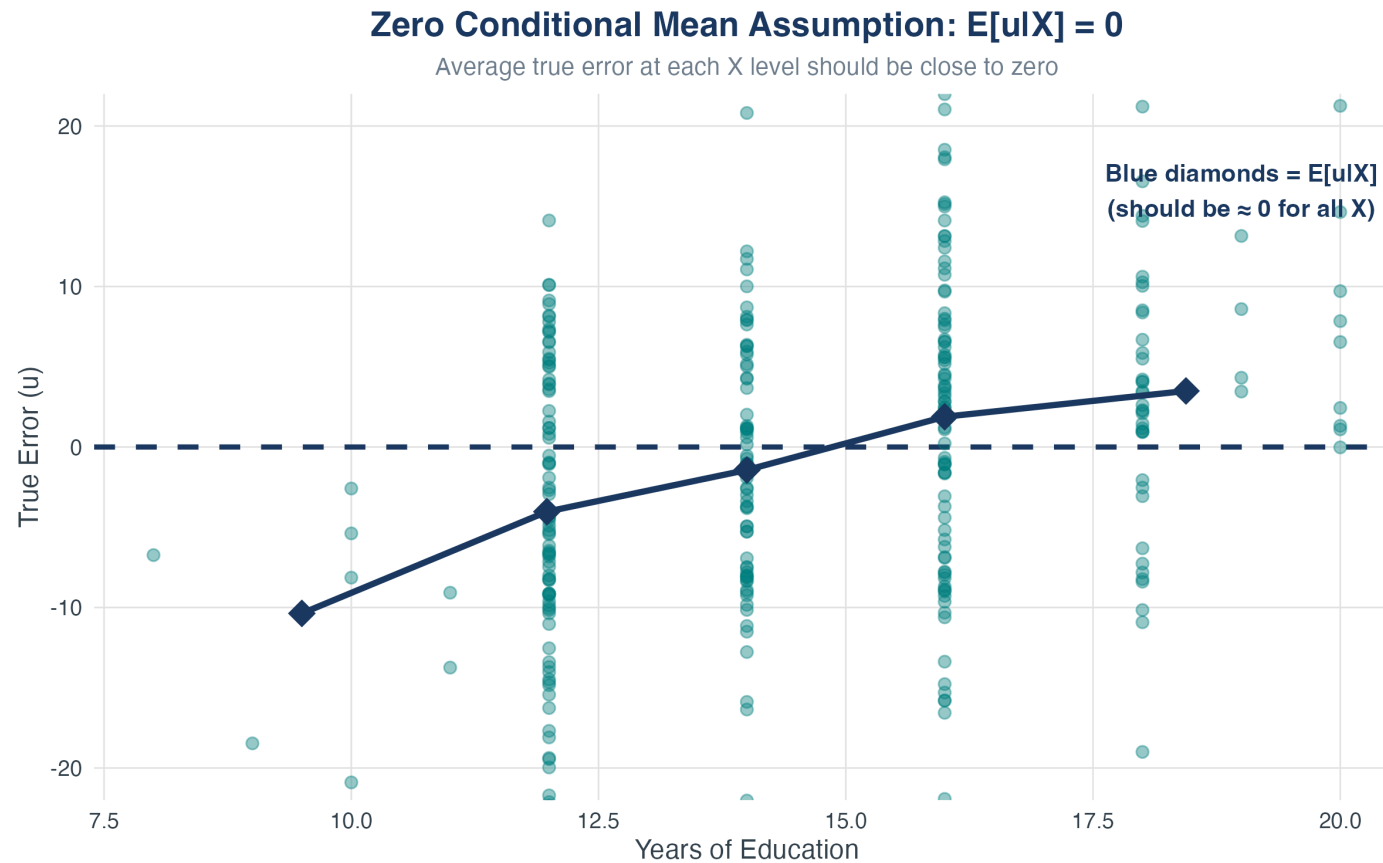
And this is the **zero conditional mean assumption**

 This is the KEY assumption

If $E[u|X] \neq 0$, then $\hat{\beta}_1$ is **biased** — it systematically estimates the wrong thing.

This is called **omitted variable bias**.

Visualizing Zero Conditional Mean



What this shows:

Residuals (teal points) are evenly scattered above and below zero

Red diamonds show the **average residual** at each education level: $E[u|X]$

Zero conditional mean assumption

Because the expected value is a linear operator, then $E[u|X] = 0$ implies that

$$E[Y|X] = \beta_0 + \beta_1 X + E[u|X] = \beta_0 + \beta_1 X$$

This shows that the **population regression function** is a linear function of X !

Nice.

When Does Zero Conditional Mean Fail?

Example: Education and wages

$$\text{Wage} = \beta_0 + \beta_1 \cdot \text{Education} + u$$

What's in u ?

Ability, motivation, family background, connections

Problem:

High-ability people tend to get more education

So $E[u|\text{Education} = \text{high}] > E[u|\text{Education} = \text{low}]$

Violates zero conditional mean!

Result: $\hat{\beta}_1$ **overestimates** the causal effect of education (it picks up both education *and* ability)

Omitted Variable Bias (OVB)

Sampling Distributions and Uncertainty

$\hat{\beta}_1$ is a Random Variable

Key insight: Our estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ depend on the sample.

Different samples $\Rightarrow$ different estimates

If we drew a new sample, we'd get a different $\hat{\beta}_1$

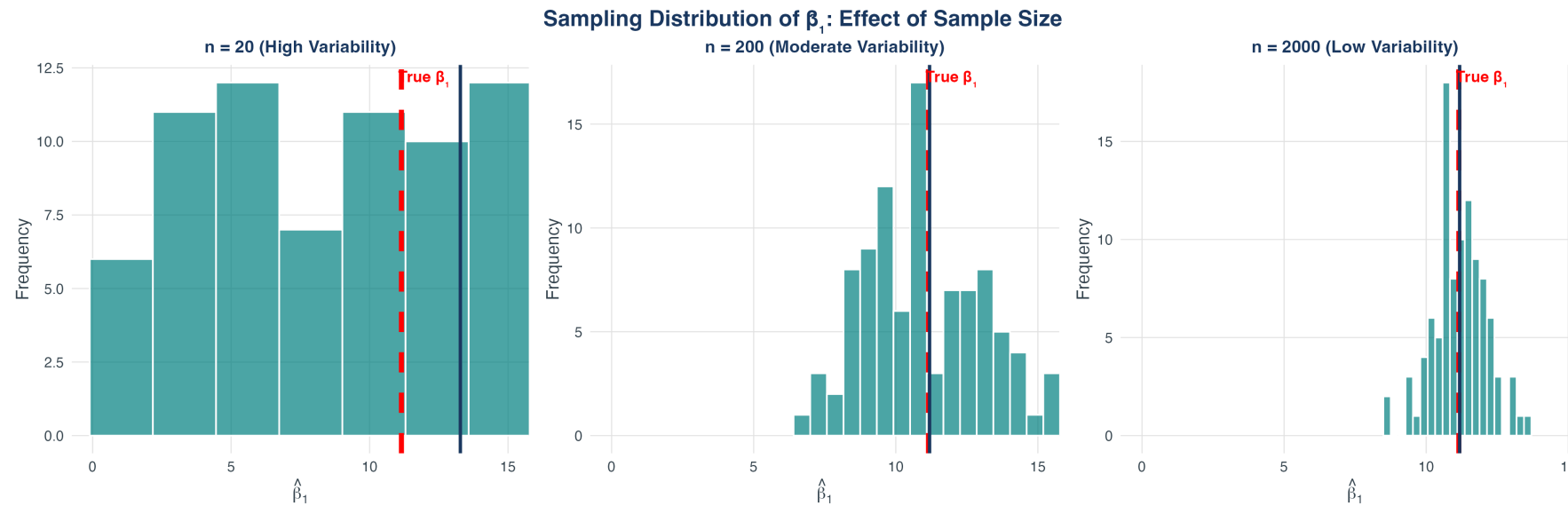
This creates a **sampling distribution**

Under the three assumptions:

$\hat{\beta}_1$ is **unbiased**: $E[\hat{\beta}_1] = \beta_1$

$\hat{\beta}_1$ is **consistent**: As $n \rightarrow \infty$, $\hat{\beta}_1 \rightarrow \beta_1$

Sampling Distribution Visualization



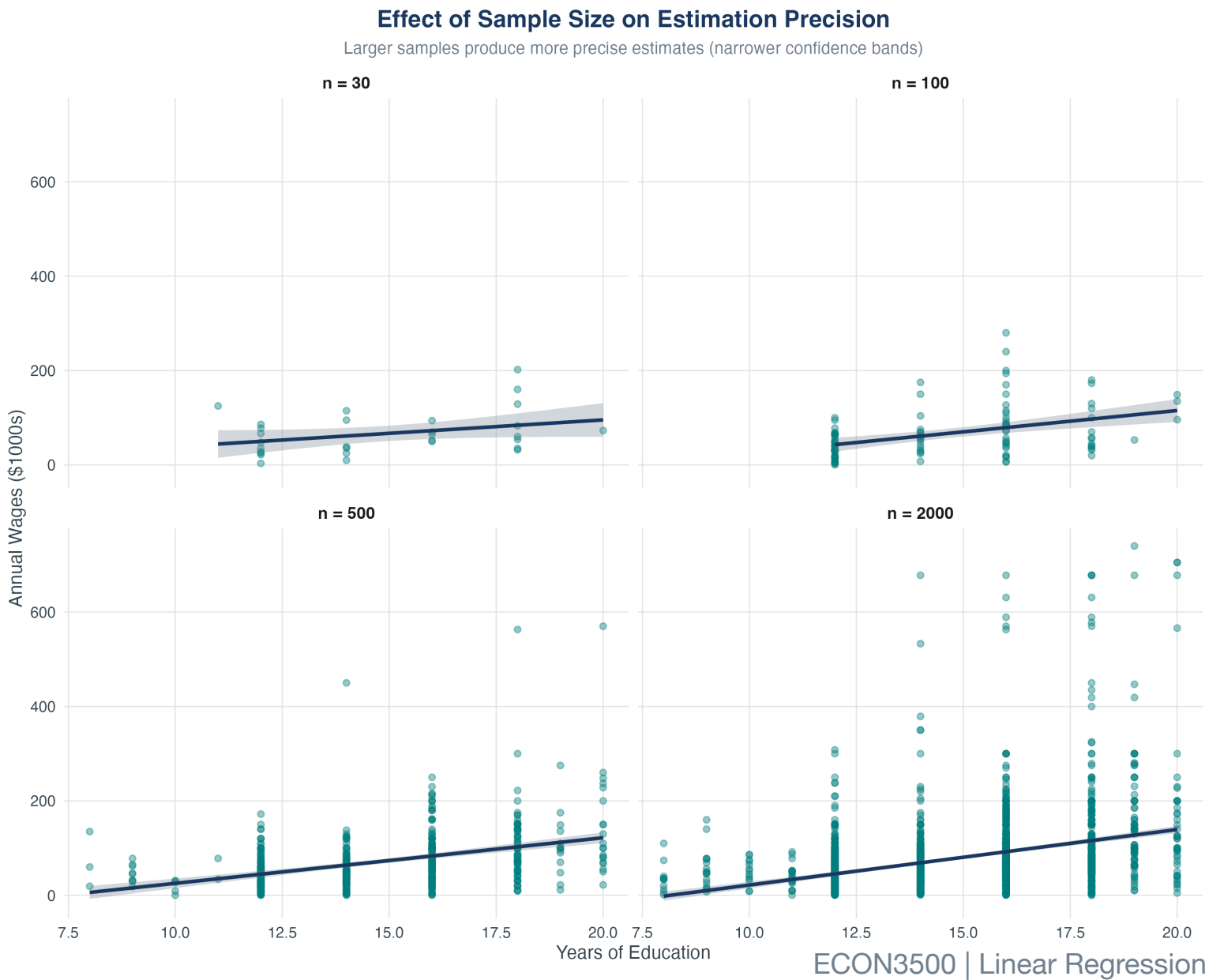
1,000 different samples, each with $n = 300$

Each sample gives a different $\hat{\beta}_1$

Mean of estimates equals true β_1 (unbiased!)

Larger $n \Rightarrow$ tighter distribution (more precise)

Effect of Sample Size on Precision



What Determines the Variance of $\hat{\beta}_1$?

The variance of $\hat{\beta}_1$ is:

$$\sigma_{\hat{\beta}_1}^2 = \frac{1}{n} \frac{\text{Var}[(X_i - \mu_X)u_i]}{\text{Var}(X_i)^2}$$

Interpretation:

Larger n : More data $\Rightarrow$ smaller variance (more precise)

Larger $\text{Var}(X)$: More spread in X $\Rightarrow$ smaller variance (easier to detect slope)

Larger σ_u^2 : Noisier errors $\Rightarrow$ larger variance (harder to estimate)

► We'll use this in the next chapter for hypothesis testing and confidence intervals.

Bringing It All Together

The Big Picture

! Key Takeaways

- . We minimize the sum of squared residuations to estimate β_1 and β_0
- . Slopes have precise interpretations: ΔY per unit ΔX , holding u fixed
- . R^2 and SER measure model fit, but don't relate to causality
- . Zero conditional mean is THE critical assumption for unbiased estimation
- . $\hat{\beta}_1$ has a sampling distribution — it's random, not fixed
- . Larger samples and more variation in X give more precise estimates

Next chapter: Hypothesis testing and confidence intervals — how to make formal inferences about β_1 .

Bonus: Common Misconceptions and Clarifications

Misconception #1: “Correlation = Causation”

Misconception

“Education and wages are correlated, so education *causes* higher wages.”

Clarification

Regression shows **association**, not necessarily causation.

Could be reverse causality (higher earners invest more in education)

Could be omitted variable bias (ability affects both)

Need experimental or quasi-experimental design for causal claims

Key point: OLS estimates the **conditional expectation** $E[Y|X]$, not a causal effect.

Misconception #2: “High R^2 = Good Model”

Misconception

“ $R^2 = 0.30$ is low, so this is a bad model.”

Clarification

R^2 depends on the context!

In physics: $R^2 > 0.95$ is common (physical laws are precise)

In social sciences: $R^2 = 0.30$ can be excellent (human behavior is complex)

R^2 measures *explained variance*, not *model validity*

Better questions:

Are coefficients statistically significant?

Do they have the expected signs?

Are residuals well-behaved?

Misconception #3: “Intercept Must Be Meaningful”

Misconception

“ $\beta_0 = -114.7$ means someone with zero education earns -\$114,700? That’s impossible!”

Clarification

The intercept is often **not interpretable** in practice.

It’s the predicted Y when $X = 0$

But $X = 0$ may be outside the range of data (no one has 0 years of education)

It’s an *extrapolation*, not an actual prediction

Key point: The intercept ensures the regression line passes through $(\bar{X}, \bar{Y})$ - it’s a technical parameter, not always economically meaningful.

Misconception #4: “Residuals = Errors”

Misconception

“The residual $\hat{u}_i$ is the same as the true error u_i .”

Clarification

They are **different** (and we never observe u_i !):

Error $u_i = Y_i - (\beta_0 + \beta_1 X_i)$ (population, unknown)

Residual $\hat{u}_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)$ (sample, observed)

Relationship: Residuals *estimate* errors, but they’re not identical.

$$\hat{u}_i = u_i - (\hat{\beta}_0 - \beta_0) - (\hat{\beta}_1 - \beta_1)X_i$$

Misconception #5: “Adding X Always Helps”

Misconception

“More variables = better model, always.”

Clarification

Only add variables that **belong** in the model:

Irrelevant variables increase standard errors (reduce precision)

Can lead to multicollinearity problems

Overfitting: model fits sample noise, not population relationship

Principle: Use **theory** to guide variable selection, not just R^2 .

Bonus: Implementing OLS in Stata

Complete Stata Workflow

Full analysis in Stata:

```
* Load data
import delimited "education_wages_data.csv", clear

* Summary statistics
summarize education wages

* Scatter plot
scatter wages education

* Correlation
correlate education wages

* Run OLS regression
regress wages education

* Generate fitted values and residuals
predict wages_fitted, xb
predict residual, residuals

* Check sum of residuals
summarize residual
```

Calculating Fit Statistics in Stata (Part 1)

Calculate TSS, ESS, and SSR manually:

```
* After running: regress wages education
* Store basic stats from regression
scalar r2 = e(r2)
scalar ser = e(rmse)
scalar n = e(N)

* Calculate mean of Y
quietly sum wages
scalar y_bar = r(mean)

* Generate squared deviations
gen deviation_total = (wages - y_bar)^2
gen deviation_explained = (wages_fitted - y_bar)^2
gen deviation_residual = residual^2
```

What we're doing: Creating three variables to capture total, explained, and unexplained variation.

Calculating Fit Statistics in Stata (Part 2)

Sum up and verify the decomposition:

```
* Sum the squared deviations
quietly sum deviation_total
scalar TSS = r(sum)

quietly sum deviation_explained
scalar ESS = r(sum)

quietly sum deviation_residual
scalar SSR = r(sum)

* Verify: TSS = ESS + SSR
display "TSS = " TSS
display "ESS = " ESS
display "SSR = " SSR
display "Check: TSS - (ESS + SSR) = " (TSS - (ESS + SSR))
```

Interpretation: The check should equal zero (or very close), confirming $TSS = ESS + SSR$.

Practice & Resources

Recommended practice:

- Stock & Watson Chapter 4 odd-numbered exercises
- Beyond our lab, practice running and interpreting regressions on the ACS data
- Practice interpreting coefficients out loud

Next session: Hypothesis testing, t -statistics, and confidence intervals